



Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Information Technology

Final Examination

Year 3, Semester 1 (2023)

IT3040 - IT Project Management

Duration: 2 Hours

May/June 2023

Instructions to Candidates:

- ◆ This paper is preceded by a **10-minute** reading period. The supervisor will indicate when answering may commence.
- ◆ This paper contains **two** parts. In **part A** there are **twelve** questions and in **part B** there are **three** questions
- ◆ Use the provided template to answer the questions in **part A**. Answer the questions in **part B** in the booklet given.
- ◆ Attach the template used to answer the questions in **part A** to the answer booklet.
- ◆ Total marks for the paper is 100 (Contributes to 30% of the final grade).
- ◆ This paper contains six pages, including the cover page.
- ◆ The use of faculty approved non-programmable calculators is allowed.

Part A

(Mark a cross "X" on the correct answer in the answer sheet given in page 6)

1. Which of the following is **not** a method of selecting a project? (1 mark)
 - a. Performing net present value
 - b. Using a power/interest grid
 - c. Implementing a balanced scorecard
 - d. Focusing on broad organizational needs
2. Which of the following statements is **true** about **payback analysis**? (1 mark)
 - a. Payback analysis is a tool used to measure the profitability of a project over its entire life cycle.
 - b. Payback analysis calculates the net present value of a project.
 - c. Payback analysis is not a commonly used financial tool in project management.
 - d. Payback analysis measures the time required for a project to earn back its initial investment.
3. Select the statement that best describes a **stakeholder**? (1 mark)
 - a. Anyone who can impact/be impacted positively or negatively by the project
 - b. Anyone who is involved in the project
 - c. Anyone positively or negatively impacted by the project
 - d. Anyone who can influence the direction of the project
4. Which of the following is **true** regarding an individual in a **Daily Scrum update**? (1 mark)
 - a. What are your impediments, what are the tasks that will get delayed, what are the tasks your teammates did?
 - b. What did you do yesterday, what are you going to do today, what are your impediments?
 - c. What did you do two days ago, what are you doing right now, what are the good things in the project?
 - d. What are your impediments, what issues did you encounter, how are you going to solve the issues?
5. Which of the following statement provides the best explanation about the **learning curve theory**? (1 mark)
 - a. All skills are plotted in a curve to identify which skill is the best skill.
 - b. When items are not produced for a long time, the company can save money because there is no cost.
 - c. When many items are produced repetitively the unit cost of those items decrease.
 - d. Learning allows to understand the project, so people need to learn more.
6. Which of the following cannot be used to identify the risks in a project? (1 mark)
 - a. SWOT Analysis
 - b. Brainstorming
 - c. Interviewing
 - d. Run Chart

7. Select the statement with the most appropriate definition for **planning poker cards**? (1 mark)
- a. Ordinary cards with the four suits.
 - b. Blank cards where individuals write their estimates
 - c. Cards with the Fibonacci series or T-shirt sizes.
 - d. Cards where only the King, Queen, Jack, and Ace cards are there.
8. Which of the following is the primary purpose of a **stakeholder register**? (1 mark)
- a. To keep a list of all project stakeholders
 - b. To serve as a project directory
 - c. To record stakeholder issues on the project
 - d. To map functional requirements but to the originate in stakeholder
9. Dinesh is working as a software project manager attached to a private international bank. A significant risk that he has recently identified is the attrition of resources. As a strategy to respond to this risk, with the support from the senior management, he provides good increments to his team members. What type of risk response is Dinesh following? (1 mark)
- a. Avoid
 - b. Transfer
 - c. Mitigate
 - d. Accept
10. Which of the following includes the main principles of the **Agile manifesto**? (1 mark)
- a. Working hardware over software, tools over individuals.
 - b. Follow a plan always, have contracts to back your projects.
 - c. Individuals and interactions over working software, customer collaboration over contract negotiation.
 - d. Customer collaboration over contract negotiation, respond to change over following a plan.
11. Which of the following statements best describes **Return on Investment (ROI)**? (1 mark)
- a. The amount of money earned from an investment divided by the cost of the investment
 - b. The total profit earned on an investment
 - c. The total amount of money invested in a project
 - d. The expected return from an investment divided by the actual return
12. Which of the following statements best defines **Risk Utility**? (1 mark)
- a. The potential benefits of an opportunity
 - b. The subjective value that an individual or organization assigns to a particular level of risk
 - c. The likelihood of a risk occurring
 - d. The expected monetary value of a risk event

Part B

Question 01

(32 marks)

Consider the following case study and answer the questions given below.

You are a project manager leading a team of 15 people in developing a new software application for a client. The project has a tight deadline of 6 months, and you have a limited budget. Your team includes a mix of experienced developers and new hires who are still learning the basics.

- Explain what human resource management is and discuss its key components. (6 marks)
- What are the steps you can take to build a strong team that is motivated to meet the project's goals? (4 marks)
- Explain **four** metrics that you can use to evaluate the performance of the team members to ensure that they are meeting the project's expectations. (8 marks)
- One of your team members is struggling to keep up with the rest of the team, and it's affecting the project's progress. What steps can you take to address this issue and help the team member get back on track? (5 marks)
- What are the potential risks associated with having a team with a mix of experienced and inexperienced members? (4 marks)
- As the project manager how can you improve team collaboration and communication? (5 marks)

Question 02

(30 marks)

Assume you are the project manager of Dazzlers, a software development company. Your team has just received a new project from a client in the healthcare domain. The project involves designing and developing a web application for managing patient data and appointments. The client has a team of four stakeholders: a project manager who is responsible for overseeing the project from the client's side, a product owner who is responsible for defining the product requirements and ensuring that the project meets the client's needs, a technical architect who is responsible for overseeing the technical aspects of the project, and a user representative who will be using the application and providing feedback.

The project manager from the client's side is a highly experienced individual who has a strong vision for the project and may have different expectations from your team. The product owner is relatively new to the healthcare domain and may not be familiar with the technicalities involved in developing such an application. The technical architect from the client's side is highly skilled but may have a rigid approach to technical decisions. The user representative has limited technical knowledge but has a strong understanding of user needs and expectations.

Given these potential differences, conflicts may arise regarding project scope, technical decisions, user needs, and project timelines. As the project manager, you need to ensure effective communication and collaboration among all stakeholders to manage these potential conflicts and ensure project success.

- a) Discuss **at least four** potential conflicts that could arise between your team and other project stakeholders. **(6 marks)**
- b) Using the power/interest grid, categorize each of the stakeholders and explain the implications of their position on the project's success. Also, identify the communication strategies that you and your team would adopt for each stakeholder category to ensure effective communication and collaboration throughout the project lifecycle. Finally, suggest some measures that you would take to manage any potential conflicts that may arise among the stakeholders during the project. Provide your answer using a similar template as given below. **(24 marks)**

Stakeholder	Power/Interest	Explanation

Question 03

(26 marks)

Use the following facts to provide answers to the questions given below. Round the answers to the **nearest whole number**.

Ruwan is a project manager of a software development project. The project involves developing a new mobile application for a client. The total budget for the project is \$200,000, and the project is scheduled to last for 12 months. It is the end of the 7th month now and Emily's CPI is 1.2.

- a) Calculate the present actual cost of the project. **(4 marks)**
- b) Assuming that the cost variance experienced thus far in the project will continue, calculate how much more money would be required to complete the project. **(8 marks)**
- c) If the variance experienced so far is expected to stop, what would be the project's estimate at completion (EAC)? **(6 marks)**
- d) Using the project's budget at completion, calculate the project's To-Complete Performance Index (TCPI). **(4 marks)**
- e) What is the percentage of the project that has been completed? **(4 marks)**

END OF EXAMINATION PAPER